The Three States of Matter

Solids, liquids and gases differ because their particles are held by different strengths of attraction.

Why solids, liquids and gases behave differently

Constituent particles attract each other with **interparticle forces**. The **strength** of these forces decides the state of matter.

	Solid	Liquid	Gas
Shape	Definite	No fixed shape (takes container's)	No fixed shape
Volume	Definite	Definite	No fixed volume (fills space)
Particle attraction	Very strong	Slightly weaker	Negligible
Particle movement	Only vibrate in fixed positions	Move freely, but stay close	Move freely in all directions

Fluids: liquids and gases both flow and have no fixed shape, so they are called **fluids**.

Changing state by heating

- **Melting point —** the (minimum) temperature at which a solid melts to a liquid at atmospheric pressure. (Ice 0°C, Iron 1538°C.)
- **Boiling point —** the temperature at which a liquid boils into vapour at atmospheric pressure (bubbles form throughout).
- **Evaporation —** slow change to vapour at the surface, at *any* temperature below boiling.

Stronger forces → higher melting point. Iron (very strong forces) melts at 1538°C; ice at 0°C.

Where students lose marks

Trap 1 — Liquids have a definite VOLUME but no definite SHAPE. They take the container's shape but keep the same volume (200 mL stays 200 mL).

Trap 2 — Boiling vs evaporation. Boiling happens at the boiling point, throughout the liquid (bubbles); evaporation is slow, at the surface, at any temperature.

Trap 3 — Gas attractions are negligible, so gases have neither fixed shape nor fixed volume and fill the whole container.

Both liquids & gases are fluids — they flow.

Model answers — write them like this

Example A. Why do liquids take the shape of their container but keep the same volume?

Step 1: Liquid particles can move freely, so the liquid has no fixed shape.

Step 2: The particles stay close together, so the volume stays the same.

∴ No fixed shape but a definite volume.

Example B. Why does iron have a much higher melting point than ice?

Step 1: Melting point depends on the strength of interparticle forces.

Step 2: Iron's forces are much stronger than ice's.

∴ Stronger forces need more heat, so iron melts at 1538°C vs 0°C for ice.

Ready to practise? → Open the worksheet